

LINEAR ALGEBRA (MTS203)

FINAL EXAM (60 MARKS)
(SOLUTION)

CLASS No. 7500 (FALL 2023)

EXAM DATE: DEC 23, 2023

Question - 1a (5 marks)

True / False.

- (a) F (b) T (c) F (d) T (e) F (f) T
(g) T (h) F (i) F (j) F

Question - 1b (4 marks)

Given

$$A = \begin{bmatrix} I & 0 & 0 \\ B & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

where all of the sub-matrices are $n \times n$, determine the block form of A^{-1} .

Solution:

$$\left[\begin{array}{ccc|ccc} I & 0 & 0 & I & 0 & 0 \\ B & I & 0 & 0 & I & 0 \\ 0 & 0 & I & 0 & 0 & I \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} I & 0 & 0 & I & 0 & 0 \\ 0 & I & 0 & -B & I & 0 \\ 0 & 0 & I & 0 & 0 & I \end{array} \right]$$

$$\therefore A^{-1} = \begin{bmatrix} I & 0 & 0 \\ -B & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

Question - 1c (3 marks)

let $X = [2 \ 1 \ 5]^T$, and let $Y = [1 \ 2 \ 4]^T$.
calculate both outer products b/w them.

$$X Y^T = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \\ = \begin{bmatrix} 2 & 4 & 8 \\ 1 & 2 & 4 \\ 5 & 10 & 20 \end{bmatrix}$$

and

$$Y X^T = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \begin{bmatrix} 2 & 1 & 5 \end{bmatrix} \\ Y X^T = \begin{bmatrix} 2 & 1 & 5 \\ 4 & 2 & 10 \\ 8 & 4 & 20 \end{bmatrix}$$

Question - 2a (5 marks)

- (a) F (b) T (c) F (d) T (e) T
(f) T (g) F (h) T (i) T (j) T

Question - 2b (4 marks)

let A be an $n \times n$ matrix. Is it possible for $A^2 + I = 0$
in the case where n is odd? Answer the same question
in the case where n is even.

$$A^2 + I = 0 \Rightarrow A^2 = -I$$

$$\det(A) \det(A) = \det(-I) = (-1)^n \det(I)$$

when n is even, $\det(A^2) \neq (-1)^n \det(I) \Rightarrow A^2 + I = 0$ ^{not possible}

when n is odd, $\det(A^2) = (-1)^n \det(I) = -\det(I) = \det(-I)$

Question - 2c (3 marks)

Let $\bar{x}$ and $\bar{y}$ be distinct vectors in $\mathbb{R}^n$ (i.e. $x \neq y$), and let A be an $n \times n$ matrix with the property that $A\bar{x} = A\bar{y}$. Show that $\det(A) = 0$.

$$\because A\bar{x} = A\bar{y}$$

$$\Rightarrow A\bar{x} - A\bar{y} = 0$$

$$\Rightarrow A(\bar{x} - \bar{y}) = 0$$

$$\text{let } \bar{z} = \bar{x} - \bar{y} \quad \because x \neq y.$$

$$\Rightarrow A\bar{z} = \bar{0}$$

$\Rightarrow \bar{z}$ be a non-zero solution of above homogeneous system

$\therefore A$ is singular

$$\Rightarrow \det(A) = 0$$

proved

Question - 3a (5 marks)

True / False

(a) T

(b) F

(c) F

(d) F

(e) F

(f) T

(g) T

(h) F

(i) F

(j) T

Question - 3b (4 marks)

Let A be a 6×4 matrix of rank 4.

(a) What is the dimension of $N(A)$? What is the dimension of the column space $R(A)$?

$\because A$ is of rank 4

$$\therefore \dim(N(A)) = 0$$

$$\text{and } \dim(R(A)) = 4.$$

- (b) Do the column vectors of A span $\mathbb{R}^6$? Are the column vectors of A linearly independent? Write a (no more than) two sentence explanation for it.

No, column vectors of A does not span $\mathbb{R}^6$.

Yes, there are only four linearly independent vectors of $\mathbb{R}^6$
So, there you can not span $\mathbb{R}^6$.

- (c) How many solutions will the linear system $A\vec{x} = \vec{b}$ have if $\vec{b}$ is in the column space of A ?
if $\vec{b}$ is in the column space of A then $A\vec{x} = \vec{b}$ will have a unique solution.

Question - 3c (3 marks)

Let $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$ be linearly independent vectors in $\mathbb{R}^n$, and let A be a nonsingular $n \times n$ matrix.

Define $\vec{y}_i = A\vec{x}_i$ for $i = 1, \dots, k$. Show that $\vec{y}_1, \dots, \vec{y}_k$ are linearly independent.

$\because \vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$ be linearly independent
So that

$$a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_k \vec{x}_k = \vec{0}$$

$$\Rightarrow a_i's = 0$$

Now,

$$A(a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_k \vec{x}_k) = A(\vec{0})$$

$$a_1 A\vec{x}_1 + a_2 A\vec{x}_2 + \dots + a_k A\vec{x}_k = \vec{0}$$

$$a_1 \vec{y}_1 + a_2 \vec{y}_2 + \dots + a_k \vec{y}_k = \vec{0} \quad \because A\vec{x}_i = \vec{y}_i$$

$$\because a_i's = 0$$

$$\Rightarrow \vec{y}_1, \vec{y}_2, \dots, \vec{y}_k \text{ are linearly independent.}$$

Question 4a (5 marks)

True / False.

(a) F (b) T (c) T (d) F (e) T

(f) T (g) F (h) F (i) T (j) T

Question 4b (7 marks)

Let L be the linear operator on $\mathbb{R}^3$ defined by

$$L(x) = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \end{bmatrix}$$

and let $S = \text{Span}(1, 0, 1)^T$. Find the kernel of L , and determine $L(S)$.

Let $x \in \ker(L)$

$$\Rightarrow L(x) = \vec{0}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad L(x) = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{array}{l} x_2 - x_1 = 0 \\ x_3 - x_2 = 0 \\ x_3 - x_1 = 0 \end{array} \right\} \Rightarrow \left. \begin{array}{l} x_2 = x_1 \\ x_3 = x_2 \\ x_3 = x_1 \end{array} \right\} \Rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\ker(L) = \text{Span}(1, 1, 1)^T \quad \underline{\text{Ans}}$$

$$L(S) = L\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{pmatrix} 0-1 \\ 1-0 \\ 1-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$L(S) = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \quad \underline{\text{Ans}}$$

Question - 5a (5 marks)

True / False

- (a) F (b) F (c) F (d) F (e) T
 (f) T (g) T (h) F (i) F (j) T

Question - 5b (7 marks)

For $A\hat{x} = b$ find all least squares solutions and show that each of these $\hat{x}$ may be written as a sum of two vectors $\bar{y}$ and $\bar{z}$ such that $\bar{y} \in \mathcal{R}(A^T A)$ and $\bar{z} \in \mathcal{N}(A^T A)$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

We immediately see that A is not full-rank.

$\therefore (A^T A)$ is singular, and consequently $(A^T A)\hat{x} = A^T b$ will have infinitely many solutions.

($\because$ it cannot be inconsistent)

$$\begin{aligned} A^T A &= \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix} & A^T b &= \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 12 \\ 12 & 24 \end{bmatrix} & A^T b &= \begin{bmatrix} 6 \\ 12 \end{bmatrix} \end{aligned}$$

it is very clear that the column space of $A^T A$ i.e. $\mathcal{R}(A^T A) = \text{span}([1 \ 2]^T)$.

Also $\mathcal{N}(A^T A)$ is:

$$\begin{bmatrix} 6 & 12 & 6 \\ 12 & 24 & 12 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 1 & 2 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

$$x_1 + 2x_2 = 1 \Rightarrow x_1 = 1 - 2x_2.$$

$$\text{let } x_2 = \alpha.$$

$$\hat{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$\nwarrow \quad \nearrow$
 $\in \mathcal{R}((A^T A)^T) \quad \in \mathcal{N}(A^T A)$

However not completely.

$$\text{So, } [1, 0]^T = \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{2}{5} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$\nearrow \in \mathcal{R}((A^T A)^T) \quad \nwarrow \in \mathcal{N}(A^T A)$

$$\therefore \mathcal{R}((A^T A)^T) = \text{span}([1, 2]^T)$$

and

$$\mathcal{N}(A^T A) = \text{span}[-2, 1]^T$$

So

$$\hat{x} = \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{2}{5} \begin{bmatrix} -2 \\ 1 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\hat{x} = \underbrace{\frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\in \mathcal{R}((A^T A)^T)} + \underbrace{(\alpha - 2/5) \begin{bmatrix} -2 \\ 1 \end{bmatrix}}_{\in \mathcal{N}(A^T A)}$$

Ans

Question - 6a (5 marks)

True/False

- (a) T (b) F (c) T (d) F (e) F
 (f) F (g) T (h) T (i) F (j) T

Question - 6b (7 marks)

let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 2 & -2 \end{bmatrix}$

Find all eigenvalues of A . For each eigenvalue, find a basis for the corresponding eigenspace.

For Eigenvalues:

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 0 \\ 1 & 1-\lambda & -1 \\ 1 & 2 & -2-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda) \begin{vmatrix} 1-\lambda & -1 \\ 2 & -2-\lambda \end{vmatrix}$$

$$= (1-\lambda) [(1-\lambda)(-2-\lambda) + 2]$$

$$= (1-\lambda) [-2 - \lambda + 2\lambda + \lambda^2 + 2]$$

$$= (1-\lambda) (\lambda^2 + \lambda)$$

$$= \lambda (1-\lambda) (1+\lambda)$$

$$\det(A - \lambda I) = 0$$

$$\lambda (1-\lambda) (1+\lambda) = 0$$

$$\boxed{\lambda = 0}$$

$$1-\lambda = 0$$

$$1+\lambda = 0$$

$$\Rightarrow \boxed{\lambda = 1}$$

$$\Rightarrow \boxed{\lambda = -1}$$

Eigenvalues: $\lambda = 0, 1, -1$

Ans

For Eigenvectors:

For $\lambda = 0$

$$A - 0I = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 2 & -2 \end{bmatrix}$$

To find the null space of $A - 0I$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 2 & -2 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow x_1 = 0$$

$$x_2 - x_3 = 0 \Rightarrow x_2 = \alpha$$

$$x_3 = \alpha$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ \alpha \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$[0, 1, 1]^T$ is basis of Null space of $A - (0)I$

$\therefore [0, 1, 1]^T$ is also the eigenvector corresponding to $\lambda = 0$. Ans

For $\lambda = 1$

$$A - I = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 1 & 2 & -3 \end{bmatrix}$$

To find the null space of $A - (1)I$

$$\Rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 2 & -3 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$y_1 + 2y_2 - 3y_3 = 0 \Rightarrow y_1 + 2\alpha - 3\alpha = 0 \Rightarrow y_1 = \alpha$$

$$y_2 - y_3 = 0 \Rightarrow y_2 = \alpha$$

$$y_3 = \alpha$$

$$\therefore y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} \alpha \\ \alpha \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore [1, 1, 1]^T$ is basis of Null space of $A - I$

is

$[1, 1, 1]^T$ is also the eigenvector corresponding to $\lambda = 1$

Ans

For $\lambda = -1$

$$A - (-1)I = A + I = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$

To find the null space of $A + I$

$$\Rightarrow \left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 1 & 2 & -1 & 0 \\ 1 & 2 & -1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 2 & -1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 2 & -1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$z_1 = 0$$

$$z_2 - \frac{1}{2} z_3 = 0 \Rightarrow z_2 = \frac{1}{2} \alpha$$

$$z_3 = \alpha$$

$$Z = [z_1, z_2, z_3]^T = [0, \frac{1}{2}\alpha, \alpha]^T = \alpha [0, \frac{1}{2}, 1]^T$$

$[0, \frac{1}{2}, 1]^T$ is basis of Null space of $A - (-1)I$
 $\therefore [0, \frac{1}{2}, 1]^T$ is also the eigenvector corresponding to $\lambda = -1$

Ans